

A GUIDE TO PRODUCT QUALITY STANDARDS CERAMIC PRODUCTS



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ABBREVIATIONS

<i>ANSI</i>	American National Standards Institute
<i>APLAC</i>	Asia Pacific Laboratory Accreditation Co-operation
<i>AS</i>	Australian Standards
<i>ASTM</i>	American Society for Testing and Materials
<i>BIS</i>	Bureau of Indian Standards
<i>CE</i>	Conformité Européenne
<i>CoC</i>	Certificate of Conformity
<i>CPSC</i>	Consumer Products Safety Commission
<i>CPSIA</i>	Consumer Product Safety Improvement Act
<i>CPSA</i>	Consumer Product Safety Act
<i>EC</i>	Council Regulation (Regulation of the European Parliament and Council)
<i>EFSA</i>	European Food Safety Authority
<i>EN</i>	European Standards
<i>EU</i>	European Union
<i>FCM</i>	Food Contact Materials
<i>FDA</i>	Food and Drug Administration
<i>FHSA</i>	Federal Hazardous Substances Act
<i>FSSAI</i>	Food Safety and Standards Authority of India
<i>GCC</i>	Gulf Co-operation Council
<i>GSO</i>	GCC Standardization Organization
<i>ILAC</i>	International Laboratory Accreditation Co-operation
<i>IS</i>	Indian Standard
<i>ISI</i>	Indian Standards Institution
<i>ISO</i>	International Organization for Standardization
<i>MLA</i>	Multilateral Recognition Arrangement
<i>MRA</i>	Mutual Recognition Arrangement
<i>MSME</i>	Micro, Small & Medium Enterprises
<i>NABCB</i>	National Accreditation Board for Certification Bodies
<i>NABL</i>	National Accreditation Board for Testing and Calibration Laboratories
<i>QCI</i>	Quality Council of India
<i>QCO</i>	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2,40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This part of the compendium brings relevant regulations and standards pertaining to the Ceramics industry and goods. Uttar Pradesh is renowned for its varied ceramic work of great aesthetic value with Bulandshahr, Azamgarh and Rampur known specially for their ceramics and pottery work.

CERAMIC PRODUCTS, BULANDSHEHER & GORAKHPUR

1. INTRODUCTION

A ceramic can be any of the various hard, brittle, heat-resistant and corrosion-resistant materials made by shaping and then firing a non-metallic mineral such as clay, at a high temperature. Common examples are earthenware, stoneware and porcelain.



Most often, fired ceramics are either vitrified or semi-vitrified. Human beings appear to have been making their own ceramics for at least 26,000 years, subjecting clay and silica to intense heat to fuse together and form ceramic materials. These however were religious figurines rather than pottery. While actual pottery fragments have been found which are up to 19,000 years old, it was not until about ten thousand years later that regular pottery became common. The earliest known pottery was made by mixing animal products with clay and baking in kilns at up to 800°C.

The invention of the wheel eventually led to the production of smoother, more even pottery using the wheel-forming technique, like the pottery wheel. Early ceramics were porous, absorbing water easily. Ceramic

pottery only became useful for more items with the discovery of glazing techniques, coating pottery with silicon, bone ash or other materials that could melt and reform into a glassy surface, making a vessel impervious to water.

Ceramics are defined as a class of inorganic, non-metallic solids, which have been subjected to high temperature during manufacturing. Most common ceramics are composed of oxides, carbides and nitrides; apart from these, Silicates and borides are also used to produce ceramics. Traditional ceramic refers to products which are produced from a combination of refined clay and powdered minerals. The general classification of traditional ceramics includes pottery and white ware. Pottery is used as a generic term for ceramics that contain clay. Pottery is not used for structural /technical or refractory purpose.

Bulandsheher and Gorakhpur are important centres of Pottery and Ceramic wares in Uttar Pradesh. The crafts date back to the times of the Tughlaq dynasty and the artisans are renowned for their creativity in incorporating various themes from day to day life into their art. There are many units engaged in this craft. The terracotta products from Gorakhpur having bagged the coveted GI tag which is a homage to the mastery of the artisans.

For the purpose of this compendium, we shall be referring to regular day use

ceramic wares and not advanced ceramics the likes of which are used for the manufacture of bullet proof armour etc.

2. RAW MATERIALS

Traditional ceramic raw materials include clay minerals such as kaolinite whereas more recent materials include aluminium oxide, more commonly known as alumina. Raw materials for ceramics span a fairly broad range due to the fact that ceramics themselves are a diverse group of materials. Generally, ceramics share the physical properties of being crystalline in structure and acting as heat and electrical insulating materials. Earthenware, pottery and bricks contain the raw materials for ceramics in the form of crystalline metals like aluminium mixed with silicate compounds like quartz, feldspar and mica which are composed largely of silicon dioxide. These types of raw materials are generally referred to as clay, constituting over 90% of the mineral compounds found on the surface of earth, that form into rocks. The modern ceramic materials which are classified as advanced ceramics include silicon carbide and tungsten carbide.

The basic raw materials required for preparation of ceramics are:

- Clay
- Feldspar, Silica, Flint (A hard Quartz)
- Ash (From Animal Bones for manufacture of Bone China)

3. MANUFACTURING PROCESS

There are three main types of pottery/ceramic ware

- Stoneware
- Porcelain
- Earthenware

The basic steps for manufacturing of all ceramics can be laid down as:

3.1. Beneficiation

It is a process that improves the economic value of the ore by removing waste materials. The clay obtained is cleaned and all visible impurities are removed.

3.2. Mixing

Mixing can be done manually in small batches or in a clay mixer if the batch size is large. This process involves mixing the clay with water till it reaches a consistency suitable for use.

3.3. Ceramic Forming

Ceramic forming is a technique of forming ceramics which are used to make different products; from tableware such as teapots to other products such as bowls, salt shakers etc.

Pottery techniques include the potter's wheel, slip casting and many others.

3.4. Green Machining

Green ceramic machining is an alternative method for the mould-free fabrication of complex-shaped ceramics where ceramics in 'unsintered' green state are

machined to the desired shape using CAD/CAM technology.

3.5. Drying

Drying is a critical step in manufacturing of ceramics as unevenly dried products can develop cracks and get damaged. Care needs to be taken to ensure the product is dried evenly.

3.6. Pre-Sintering

Sintering or firing is a process by which the ceramics are thermally consolidated into a dense cohesive body comprising of fine uniform grains. This process is also called densification. This process is carried out in two stages:

- a. **Bisque Firing:** It is done to vitrify the green ware which means, “to turn it glasslike,” to a point that the pottery can have a glaze adhering to the surface.
- b. **Glaze Firing:** Ceramic glaze is an impervious layer or coating applied to bisque ware to colour, decorate or waterproof an item. For earthenware, such as fired clay pottery, to hold liquid, a glaze is required.

3.7. Glazing

Ceramic glaze is an impervious layer or coating of a vitreous substance which has been fused to a ceramic body through firing. Glaze can serve to colour, decorate or waterproof an item. Glazing renders earthenware vessels suitable for

holding liquids, sealing the inherent porosity of unglazed biscuit (or Bisque) earthenware. Glaze also gives a tougher surface and used on stoneware and porcelain as well.

3.8. Sintering

This is the final step in the process where the glaze fuses with the ceramic ware giving it its characteristic look.

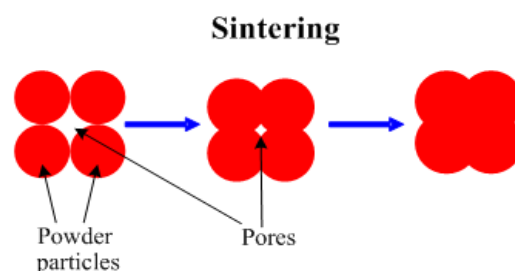


Figure 1: Sintering process

Ceramics can be classified into different categories, based on the temperatures they are fired in the kiln:

A. STONEWARE (CERAMICS)

Stoneware pottery is a vitreous or semi-vitreous ceramic made from non-refractory fire clay. Stoneware is known for its scratch resistance and non-porous texture; stoneware pottery is usually grey or brownish in colour. The basic steps involved in the manufacturing of stoneware pottery are:

- a. **Mixing the clay:** Using a flat surface, the clay is kneaded well. It is periodically cut with special knives to remove air bubbles.
- b. **Shaping:** The well kneaded clay is shaped into the desired form manually with hands or by using a potter's wheel. On completion, it is placed to dry.

- c. **Glazing:** Glaze is applied using either a dip method where pots are dipped in a large container of glaze or by brushing.
- d. **Firing in the Kiln:** In this process, the stoneware is fired at temperatures between about 1100 °C (2010 °F) to 1300 °C (2370 °F). This causes the product to attain its characteristic appearance.



Figure 2: Kiln Firing



Figure 3: Characteristic Stoneware

B. PORCELAIN (CERAMICS)

Porcelain can be described as a ceramic material, including materials like 'Kaolinite' in a furnace to temperatures between 1,200 and 1,400 °C (2,200 and 2,600 °F). It originated from China around 1,200 years ago and is often referred to as China. Porcelain is broadly classified into three

categories depending on the composition of the paste which is employed to make the body of the porcelain object.

- **Hard paste porcelain:** Chiefly refers to formulations made from mixtures of kaolin, quartz and feldspar which are fired at temperatures of around 1400 °C.
- **Soft paste porcelain:** Chiefly refers to formulations made from mixtures of ball clay, ground glass, bone ash, soapstone, flint and quartz which are fired at temperatures of around 1200 °C.
- **Bone China:** Chiefly refers to formulations made from kaolin, feldspathic materials and bone ash (a minimum of 30% phosphate from animal bones). This is the strongest of all porcelains and is known for its high levels of translucency. This strength allows it to be fashioned into articles of thin cross sections and provides an overall dainty look.

The basic steps involved in the manufacturing of porcelain are:

- a. **Mixing and crushing the raw materials:** First, the raw material particles are reduced to the desired size which involves using a variety of equipment during several crushing and grinding steps for fine grinding for which ball mills are used that consist of large rotating cylinders partially filled with steel or ceramic grinding media of spherical shape.
- b. **Cleaning and Mixing:** The ingredients are passed through a series of screens to remove any under- or over-sized

materials. If the body is to be formed wet, the ingredients are combined with water to produce the desired consistency. Magnetic filtration is then used to remove iron from the slurries as iron imparts an undesirable reddish hue to the body if it oxidizes during firing. If the body is to be formed dry, shell mixers, ribbon mixers or intensive mixers are typically used.



Figure 4: Ball Mill Mixers

- c. **Shaping:** The desired shapes are given to the raw materials by either manual throwing, moulding or jiggering.
- d. **Firing in the Kiln:** The formed products are subjected to a bisque firing which can be called as a first firing to vaporize volatile contaminants.
- e. **Glazing:** Glaze is then applied to the ware by means of painting, pouring, dipping or spraying. Different types of glazes can be produced by varying the proportions of the constituent ingredients such as alumina, silica, and calcia; for example, increasing the alumina and decreasing the silica produces a matte glaze.
- f. **Firing:** The products are fired in a kiln at temperatures between about 1200 °C

(2190 °F) to 1400 °C (2550 °F) to obtain the final product.



Figure 5: Automatic porcelain shaping machine

For manufacturing of Bone China (a type of Porcelain), ground animal bone ash is mixed with the raw materials to produce porcelain which has a translucent appearance. It is also decorated extensively by spraying, decal applications etc.

After application of the decoration, Bone China is fired once again in the kiln to fuse the pigments into the base.

C. EARTHENWARE (TERRACOTTA)

Earthenware are usually glazed or unglazed non-vitreous pottery which have been normally fired at temperatures below 1,200 °C (2,190 °F). Earthenware are usually made from common clay which is a low fire clay for use in pottery and sculptures. It contains a lot of minerals such as iron oxide (rust) and in its raw state may contain some sand or small bits of rock. The basic steps involved in manufacturing are:

- a. **Mixing the clay:** Using a flat surface, the clay is kneaded well. It is periodically cut, using special knives to remove air bubbles.

- b. Shaping:** Manually with hands or by using a potter's wheel. On completion, it is placed to dry.



Figure 6: Shaping the Earthenware

- c. Firing in the Kiln:** Earthenware is fired at temperatures in the range of about 1000°C (1830 °F) to 1200 °C (2190 °F). The earthenware produced, is still porous unlike stoneware, porcelain or Bone China and has to be made impervious to water if to be used for holding liquids by application of a slip or a glaze.
- d. Application of Slip:** Earthenware is slightly porous after the first firing. It is made waterproof by the application of slip (which is a liquid clay mixture) or a glaze, before the second firing to render it impervious to water.

Earthenware is considered to be a type of Terracotta. The most easily identifying features of terracotta is the reddish brown, earthen colours of the finished products though many other colours may be added.



Figure 7: Finished earthenware

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for warpage (deforming)
- Check for shapes and dimensions
- Check for uniformity in glazing and any other visual defects such as cracks, chipping etc.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (<https://www.fieo.org/>)

- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade

Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce

(<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic (for food contact vessels only)

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	a. Food Safety and Standards (Licensing and Registration of Food Businesses) Regulations, 2011	a. Food Safety and Standards Authority of India (FSSAI)	a. Governs use of food grade utensils b. All equipment and utensils which come in contact with food.

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Commission Regulation (EC) No. 2023/2006 of 22 December 2006, on good manufacturing practice for materials and articles intended to come into contact with food. b. Framework Regulation (EC) No. 1935/2004	a. European Commission	a. Specifies requirements with respect to Chemicals and heavy metals b. Specifies Good manufacturing practices for materials coming in contact with food c. Specifies General safety requirements for all food contact materials. In addition, specific measures have been introduced to plastics
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008 b. Consumer Product Safety Act (CPSA), Act of 1972 c. Federal Hazardous Substances Act (FHSA), Act of 1960. d. California Proposition 65, Act of 1986. e. Code of Federal Regulations Title 21, Sec.109.16	a. California State b. Consumer Products Safety Commission c. Food and Drug Agency (FDA)	a. Specifies requirements with respect to Chemicals and heavy metals b. Specifies requirement for suitable Product Labelling c. Specifies Banned substances d. Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products e. Regulations to monitor Lead content in glazes, ornamental ceramics
3	Middle East	a. GSO Technical Regulations ▪ GSO 2231: 2012 ▪ GSO 839: 2001	a. GCC Standardization Organization (GSO)	a. General requirements for the materials intended to come into contact with food

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
General (Refer Table 1)	IS 2781: 2012	Glossary of terms relating to ceramic ware (third revision)	USA	ASTM C242-01	Standard terminology of ceramic white wares
	-	-	Australia	AS 4371-2012	Ceramic tableware specifications
	-	-	Middle East	GSO 839: 2001	Food packages – Part 1: General requirements
	-	-	Middle East	GSO 2231: 2012	General requirements for materials intended to come into contact with food
Raw Materials (Refer Table 2.a & 2.b)	IS 2333: 1992 (Reaffirmed Year: 2019)	Plaster of Paris for ceramic industry	USA	ASTM C322-09	Standard practices for sampling white ware clay
	IS 2840: 2021	China clay for ceramic industry	USA	ASTM C326-09 (2018)	Standard test method for testing drying and firing shrinkages of ceramic white ware clays
	IS 4589: 2002 (Reaffirmed Year: 2018)	Plastic clay and washed plastic clay for ceramic industry	Middle East	GSO ISO 14720-1: 2016	Testing of ceramic raw and basic materials - Determination of sulphur in powders and granules of non-oxide ceramic raw and basic materials
	-	-		ISO 14720 - 1: 2013	Testing of ceramic raw and basic materials — Determination of sulphur in powders and granules of non-oxide ceramic raw and basic materials — Part 1: Infrared measurement methods.
Manufacturing/ Processing (Refer Table 3a. & 3.b)	IS 2838: 1964 (Reaffirmed Year: 2019)	Stoneware containers for general purposes	European Union	BS EN 12983-1-2000	Domestic cooking ceramic for use on stove
	IS 2857: 2002 (Reaffirmed Year: 2018)	Earthenware crockery ware	European Union	BS EN 14350: 2020	Childcare articles
	IS 3505: 2002 (Reaffirmed Year: 2018)	Porcelain crockery ware	Middle East	GSO CEN/TS 12983- 2: 2012	Cookware - Domestic cookware for use on top of a stove
	IS 11475: 2002 (Reaffirmed Year: 2018)	Stoneware crockery ware	-	-	-
Finishing (Refer Table 4.a & 4.b)	IS 9806: 2001 (Reaffirmed Year: 2017)	Methods of test for and permissible limits of toxic materials released from ceramic ware in contact with food	European Union	BS EN 12875-5: 2006	Mechanical dishwashing resistance of utensils - Part 4: Rapid test for domestic ceramic articles

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
	IS 14179: 1999 (Reaffirmed Year: 2019)	Methods of test for ceramic tableware	USA	ASTM C674-13 (2018)	Standard test methods for flexural properties of ceramic white ware material
	-	-	USA	ASTM C738 - 94(2020)	Standard Test Method for Lead and Cadmium Extracted from Glazed Ceramic Surfaces
	-	-	Australia	ISO 8391-2: 1986	Ceramic cookware in contact with food - Release of lead and cadmium
	-	-	Middle East	GSO ISO 6486-1: 2016	Ceramic ware, glass-ceramic ware and glass dinnerware in contact with food - Release of lead and cadmium

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 2781: 2012 - Glossary of terms related to ceramic ware	Glossary of terms related to ceramics
	AS 4371-2012 - Ceramic tableware	Specification for ceramic and China tableware
	ASTM C242-01 - Standard terminology of ceramic white wares and related products	- Glossary of terms - Referenced documents
	GSO 839: 2001 - Food packages – Part 1: General requirements	- Governs packaging of food materials - Defines primary and secondary packaging - Materials used in fabricating, forming, or treating packages for food must be “food grade,” defined as a material that does not cause any hazards or health problems, and must be clean so as to avoid contamination of food
	GSO 2231: 2012 - General requirements for materials intended to come into contact with food	- Labelling regulations - Regulates materials and articles that contact, are intended to contact, or are reasonably expected to contact food - Exclusions

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 2333: 1992 (Reaffirmed Year: 2019) - Plaster of Paris for ceramics industry, specification	- Methods of sampling and tests - Documents referenced - Physical and chemical requirements for Plaster of Paris
	IS 2840: 2021 - China clay for ceramic industry-Specification	- Methods of sampling and testing - Documents referenced - Requirements for China clay for ceramic industry
	IS 4589: 2002 (Reaffirmed Year: 2018) - Plastic and washed plastic clay for ceramic industry	- Methods of sampling and tests - Grades of materials and description - Documents referenced - Physical and chemical requirements for plastic and washed plastic clays

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	ASTM C322-09 - Standard practice for sampling ceramic white ware clays	- Sampling of bulk and bagged ceramic white ware clays - Possibility of segregation and variability of the ceramic powder in containers of the various sizes and numbers - Units of measurement specification
	ASTM C326-09 (2018) - Standard test method for drying and firing shrinkages of ceramic white ware clays	- Determination of linear shrinkage of ceramic white ware clays, fired and unfired - Units of measurement specification
	GSO ISO 14720-1: 2016 - Testing of ceramic raw and basic materials - Determination of sulphur in powders and granules of non-oxidic ceramic raw and basic materials - Part 1: Infrared measurement methods	- Determination of sulphur in powdered and granular non-oxide ceramic raw materials - Applicability of standard explained with reference to materials with mass fraction of sulphur .005% to 2% - Applicability of standards explained with reference to materials with higher mass fractions of sulphur after verification of the particular case.
	ISO 14720 - 1: 2013 - Testing of ceramic raw and basic materials — Determination of sulphur in powders and granules of non-oxide ceramic raw and basic materials — Part 1: Infrared measurement methods.	- Determination of sulphur in powdered and granular non-oxide ceramic raw materials and materials, such as silicon carbides - Applicability of standard explained with reference to materials with mass fraction of sulphur .005% to 2% - Applicability of standards explained with reference to materials with higher mass fractions of sulphur after verification of the particular case.

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 2838: 1964 (Reaffirmed Year: 2019) - Specification for stoneware containers for general purposes	- Terminology and adjunct standards - Capacities, tolerances and other requirements - Sampling and test methodology
	IS 2857: 2002 (Reaffirmed Year: 2018) - Earthen crockery ware specifications	- Terminology - Workmanship details, finish and warpage - Visual inspection guidelines
	IS 3505: 2002 (Reaffirmed Year: 2018) - Porcelain crockery ware specifications	- Terminology - Documents referenced - Types and grades defined - Classification of defects
	IS 11475: 2002 (Reaffirmed Year: 2018) - Stoneware crockery ware specifications	- Terminology - Documents referenced - Types and grades defined - Classification of defects

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	BS EN 12983-1-2000 - Cookware: Domestic cookware for use on top of a stove, cooker or hob general requirement	General requirements for cookware to be used on top of cooker, stove or hob
	BS EN 14350: 2020 - Child care articles. Drinking equipment. Safety requirements and test methods	Childcare articles, safety requirements and tests

Stage	Code	Scope
	GSO CEN/TS 12983- 2: 2012 - Cookware - Domestic cookware for use on top of a stove, cooker or hob - Part 2: Further general and specific requirements for ceramic, glass and glass ceramic cookware	General requirements for cookware to be used on top of cooker, stove or hob

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 9806: 2001 (Reaffirmed Year: 2017) - Methods of test for and permissible limits of toxic materials released from ceramic ware, vitreous enamelware, glassware and glass-ceramic ware in contact with food	- Specification of permissible limits of toxic materials released - Referenced documents - Terminology - Testing methods
	IS 14179: 1999 (Reaffirmed Year: 2019) - Methods of test for ceramic tableware	- Methods of test and sampling for ceramic tableware - Referenced documents - Number of tests and criteria for conformity

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	BS EN 12875-5: 2006 - Mechanical dishwashing resistance of utensils rapid test for domestic ceramic articles	Mechanical dishwashing resistance for domestic use ceramic
	ASTM 674-13 (2018) - Standard test methods for flexural properties of ceramic white ware materials	Determining the modulus of rupture and the modulus of elasticity, which may be required in product specifications.
	ASTM C738 - 94(2020) - Standard test method for lead and cadmium extracted from glazed ceramic surfaces	- Precise determination of lead & cadmium extracted from glazed ceramic surfaces by acetic acid - Exclusions - Specification of SI units
	ISO 8391- 2: 1986 - Ceramic cookware in contact with food — Release of lead and cadmium — Part 2: Permissible limits	Permissible limits for the release of lead and cadmium by ceramic cookware intended for use in contact with food
	GSO ISO 6486-1: 2016 - Ceramic ware, glass-ceramic ware and glass dinnerware in contact with food - Release of lead and cadmium - Part 1: Test method	Test method for the release of lead and cadmium from ceramic ware, glass-ceramic ware, and glass dinnerware intended to be used in contact with food.

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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<https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=111&showFR=1&subpartNode=21:2.0.1.1.11.4>
 - c. AUS: <https://www.compliancegate.com/food-contact-material-regulations-australia/>
 - d. GULF (GCC): <https://sabersaudi.co/services/saso-product-coc/saso-technical-regulation/>
10. Domestic Standards: <https://standardsbis.bsbedge.com/>
11. ISO Standards: <https://www.iso.org/obp/ui#search>
12. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
13. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials,

moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products,

services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such

products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL):** <https://nabl-india.org/>
- **Bureau of Indian Standards (BIS):** <https://bis.gov.in/>

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an

organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the

relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India

(QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
<http://nabcb.qci.org.in/>

APPENDIX II: IMPORTANT INSTITUTIONS

CERAMICS				
S. No.	Institute	Areas of Expertise	Postal Address	Contact
1	Centre for the Development of Glass Industry, Industrial Area, Jalesar Rd, Firozabad, Uttar Pradesh 283203	- Product and Technology Development Unit - Consultancy - Testing laboratories	Centre for the Development of Glass Industry, Industrial Area, Jalesar Rd, Firozabad, Uttar Pradesh 283203	Email: cdgizbd@gmail.com Phone: 9897021838 Website: http://www.cdgiindia.net/
2	Indian Institute of Technology – Banaras Hindu University (IIT - BHU), Department of Ceramic Engineering	Glass & Glass Ceramics	IIT- BHU, Ajagara, Varanasi, Uttar Pradesh 221005	Email: head.cer@iitbhu.ac.in Phone: 9936182124 Website: www.iitbhu.ac.in/dept/cer
3	CSIR-Central Glass & Ceramic Research Institute, 196, Raja S.C. Mullick Road, Kolkata – 700 032 West Bengal, India	- Non-oxide Ceramics - Consultancy - Refractory and Traditional Ceramics - Specialty Glass	CSIR-Central Glass & Ceramic Research Institute, 196, Raja S.C. Mullick Road, Kolkata – 700 032 West Bengal, India	Email: dir_office@cgcri.res.in Phone: +91 33 2473 0957 Website: https://www.cgcri.res.in/
4	CSIR-Central Glass & Ceramic Research Institute, CSIR-CGCRI Khurja Centre G T Road, Khurja 203131 (Uttar Pradesh)	- Consultancy - Training & Demonstration - Testing Laboratories - Trouble shooting	CSIR-Central Glass & Ceramic Research Institute, CSIR-CGCRI Khurja Centre G T Road, Khurja 203131 (Uttar Pradesh)	Email: hstripathi@cgcri.res.in Phone: +91 5738245433 Website: https://www.cgcri.res.in/campus/khurja

